University of Toronto
Faculty of Applied Science and Engineering
ECE 295
Milestone 1 (M1): Technical Design Review

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Project Title	Milestone 1
Tutorial Section	0105
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Own Work Declaration: We, Leo Stewart, Muhammad Waquas and Mayan Saravanabavan state that this report is entirely our own work and writing, and it appropriately references all sources of information, including any use of generative AI, as outlined in the academic integrity policy in the ECE295 course syllabus.

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1.0 Subsystem Overview

Subsystem F is the transmit power amplifier and lowpass filter of the Flexible Radio Transceiver (FLRTRX). It receives a low-power differential RF signal PA_IN_P and PA_IN_N from Subsystem E, amplifies it to 1-10 W, and filters the output to suppress harmonic distortion before transmission. The subsystem operates across the 8-16 MHz band, delivering its output into a 50 Ω resistive load with total harmonic distortion (THD) below 10% at 14 MHz. An active-low TXEN control signal from Subsystem C disables the amplifier during receive mode, preventing unintended transmissions and conserving power. The filtered output (PA_OUT) is routed to the mainboard's TX/RX switch for delivery to the antenna through the shared signal path.

As the final active stage before the antenna, Subsystem F's performance directly determines transmission quality. Insufficient power, excessive harmonics, or enable-control failure would degrade or prevent communication. The subsystem interfaces with Subsystem E (RF input), Subsystem C (TXEN control), and the mainboard (power rails: +5 V, +12 V, and output path). All components must fit within a 50.8 mm × 90.2 mm daughtercard.

2.0 Subsystem Design

The design implements a single-ended Class D switching amplifier followed by a fifth-order Butterworth lowpass filter with integrated impedance matching, and an RF choke (RFC). Figure A1 in Appendix A shows the complete schematic diagram with reference designators.

2.1 Input Network

A single-ended topology minimizes board area and component count compared to a push-pull alternative. PA_IN_P is AC-coupled through C1 (0.1 μF) to the voltage driver, blocking DC while passing all frequencies in the 8-16 MHz band. PA_IN_N is terminated to ground to satisfy the differential interface requirement.

2.2 Voltage Driver (U1 – LTC6752-2)

The 1 V_{pp} sinusoidal input can not directly switch a power MOSFET. The LTC6752-2 ultrafast comparator (5 V supply) converts it into a 5 V rail-to-rail pulse wave. A resistive divider (R1 = R2 = 10 kΩ) sets a 2.5 V reference at the inverting input (-IN). A second resistive divider (R3 = R4 = 10 kΩ) biases the non-inverting input (+IN) at 2.5 V, with the RF signal AC-coupled onto this node through C1. The RF signal crosses this threshold twice per cycle, producing a square wave output. The LTC6752-2 was selected for its 2.9 ns propagation delay (critical at 16 MHz)

and rail-to-rail output. The TLV3501 was considered but its 4.5 ns delay reduces edge accuracy at 16 MHz. Figure B1 in Appendix B shows the output of the voltage driver.

2.3 Enable Control (Q1 – 2N3904)

The TXEN signal controls whether the PA is active. When TXEN is low (0 V), the PA amplifies normally, and when TXEN is high (3.3 V), the PA is disabled. A 2N3904 NPN transistor (Q1), driven through base resistor R6 (10 kΩ), implements this control. The SHDN net, which connects to the shutdown/enable pins of the LTC6752-2 (U1) and MCP14A0602 (U2), is held high by a pull-up resistor R5 (10 kΩ) connected to the 5 V supply. When TXEN is high, Q1 saturates and pulls the SHDN net low, disabling the comparator and gate driver. When TXEN is low, Q1 is off and R5 pulls SHDN high, allowing signal flow. This provides sub-microsecond gating with minimal components. Figure B2 shows the output when TXEN is high.

2.4 Current Driver (U2 – MCP14A0602)

The IRF510 power MOSFET's gate capacitance (~180 pF) requires approximately 2A peak current to switch at 16 MHz. This far exceeds the comparator's 20-40 mA capability. The MCP14A0602 gate driver provides 6 A peak current, 9 ns rise / 7 ns fall time, and ~20 ns propagation delay. Powered from +12 V, it swings the gate well above the IRF510's threshold. A 1.5 Ω series gate resistor (R7) damps ringing from lead inductance. A bypass capacitor (C3 = 10 μF) adjacent to U2 supplies instantaneous switching current. Figure B3 shows the current the MCP14A0602 produces at the gate of the MOSFET.

2.5 Power Amplifier (Q2 – IRF510)

The IRF510 operates as a Class D switch in common-source configuration. The 12 V supply feeds the drain through a 10 μH RF choke (L1), which presents high impedance at RF while passing DC. AC coupling (C4 = 100 nF) passes only the RF component to the output filter. This AC-coupled topology enables drain voltage swings of ~ $2\times V_{DD}$, increasing deliverable power. The IRF510 was chosen over the BS170 (5.6 A vs. 0.5 A max drain current, 43 W vs. 0.83 W power dissipation) and the AFT05MS004N (requires SMD soldering onto a breakout board). A TO-220 heatsink manages dissipation. A 10 kΩ pull-down (R8) holds the gate low when idle.

2.6 Lowpass Filter and Impedance Matching

A fifth-order Butterworth filter (C-L-C-L-C) attenuates switching harmonics while matching the PA's source impedance to the 50 Ω load. Butterworth was chosen over Chebyshev for its

maximally flat passband, ensuring consistent power delivery across 8-16 MHz. The amplifier's effective source impedance is $R_S = V^2/(2P) \approx 7.5\ \Omega$ (at 9.5 W target). Using N = 5 prototype values ($g_1 = g_5 = 0.618$, $g_2 = g_4 = 1.618$, $g_3 = 2.0$), each element is impedance-scaled to transition from 7.5 Ω at the PA side to 50 Ω at the load. C5 (240 pF) accounts for IRF510 $C_{oss} \approx 140$ pF. L2 (360 nH) and C6 (470 pF) handle the bulk of impedance matching, while L3 (680 nH) and C7 (100 pF) provide harmonic rejection and final impedance smoothing to 50 Ω.

2.7 Simulation Results

LTspice transient simulation of the full signal chain at 14 MHz yields an RMS load voltage of 21.893 V (Figure B4) across 50 Ω ($P_{out} = V^2/R \approx 9.5861$ W, Figure B5). Fourier analysis reports THD of 1.23% at the filtered output (Figure B6). Across 8-16 MHz, power stays between 4.9W and 9.6W (Figure B7). Table C1 in Appendix C summarizes the component values used in the simulation. Overall, these results confirm the design meets all ICD specifications.

Appendix A: Schematic Design

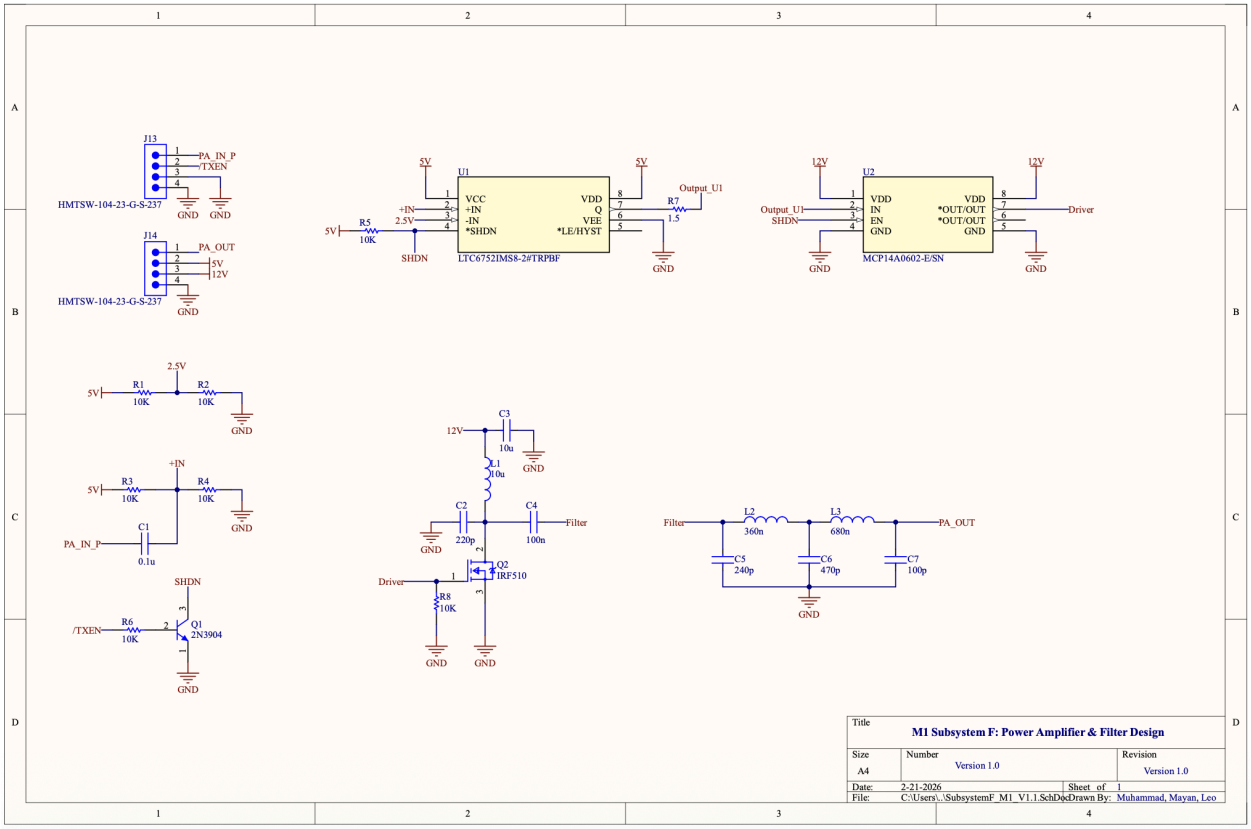


Figure A1. Subsystem F schematic diagram.

Appendix B: LTspice Simulations

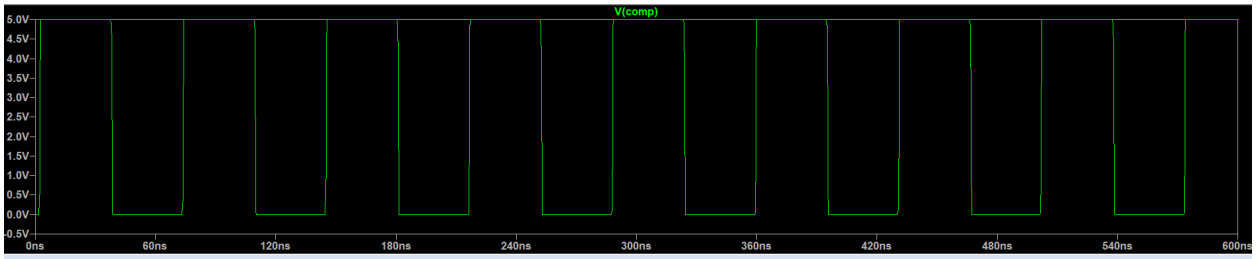


Figure B1. Output voltage of the voltage driver (LTC6752-2).

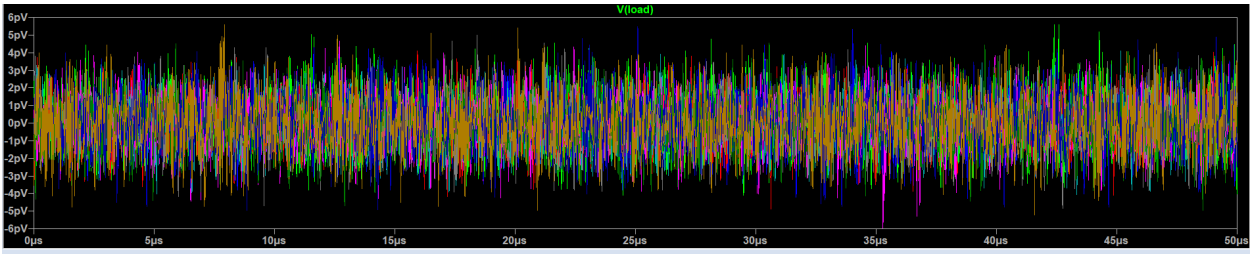


Figure B2. Output voltage of the subsystem when TXEN is HIGH. The output voltage varies between -5 to 5pV which is negligible.

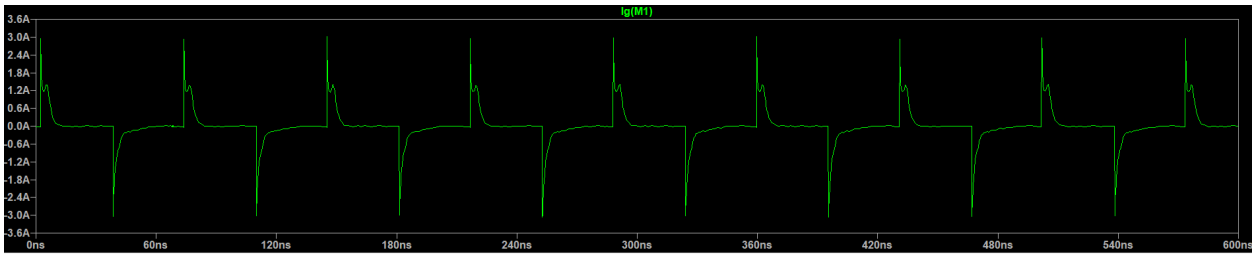


Figure B3. Current produced by the current driver (MCP14A0602) at the gate of the MOSFET. The sharp positive and negative spikes are the charging and discharging of the MOSFET's internal gate capacitance. The peak at 1.2A is because of the Miller Effect.

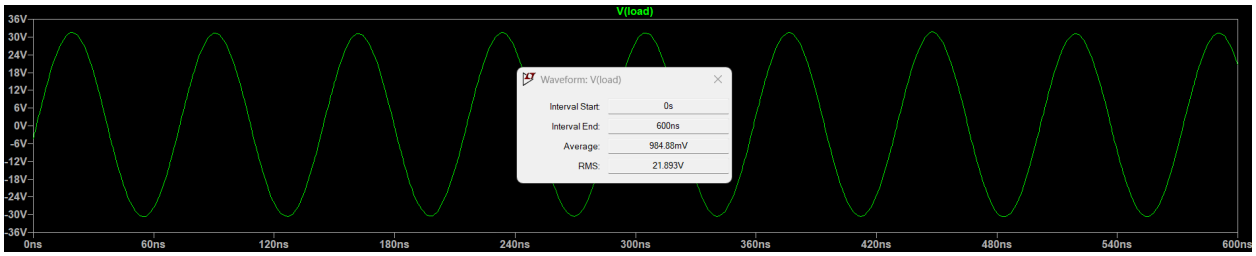


Figure B4. Shows the output voltage of the subsystem at 14 MHz. The window also shows the RMS value of the voltage which is 21.893V.

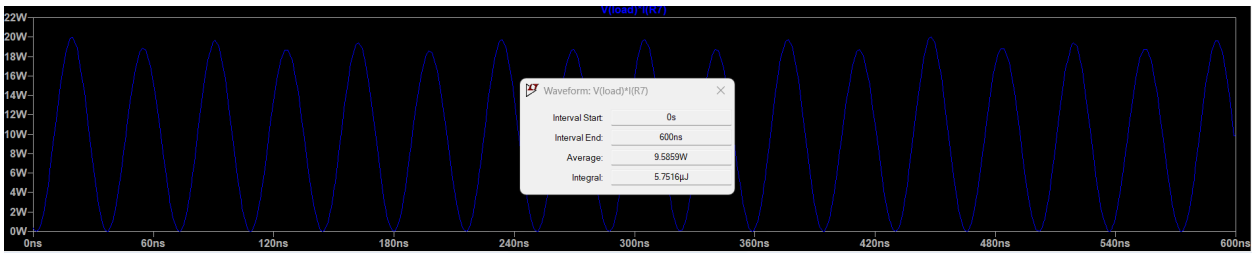


Figure B5. Shows the output power of the subsystem at 14MHz. The window also shows the average power which is 9.5859W.

```
Fourier components of V(load)
N-Period=1
DC component:-0.0123849

Harmonic     Frequency     Fourier     Normalized    Phase       Normalized
 Number        [Hz]       Component    Component   [degree]     Phase [deg]
    1        1.400e+07    3.077e+01    1.000e+00     97.30Â°       0.00Â°
    2        2.800e+07    3.654e-01    1.188e-02   -158.56Â°    -255.86Â°
    3        4.200e+07    3.827e-02    1.244e-03   -100.05Â°    -197.35Â°
    4        5.600e+07    3.880e-02    1.261e-03    -27.90Â°    -125.20Â°
    5        7.000e+07    3.105e-02    1.009e-03    152.22Â°      54.92Â°
    6        8.400e+07    2.513e-02    8.166e-04    -86.96Â°    -184.26Â°
    7        9.800e+07    2.630e-02    8.548e-04    -25.90Â°    -123.20Â°
    8        1.120e+08    1.736e-03    5.641e-05   -154.17Â°    -251.47Â°
    9        1.260e+08    1.539e-02    5.003e-04    137.19Â°      39.89Â°
Partial Harmonic Distortion: 1.211750%
Total Harmonic Distortion:   1.232133%
```

Figure B6. Shows the Fourier analysis at 14MHz. The THD is recorded at the bottom and is 1.23%.

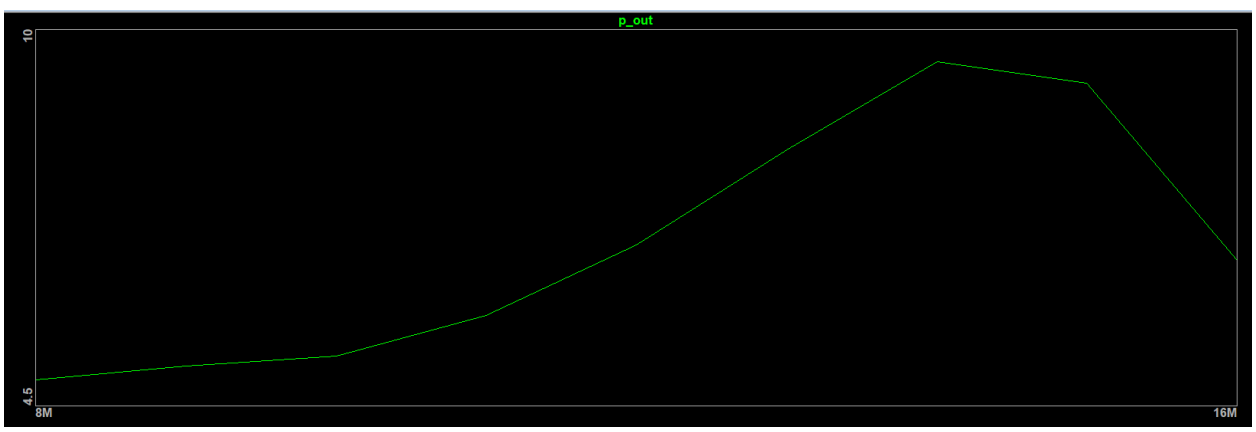


Figure B7. Shows the output power for input frequencies between 8-16MHz.

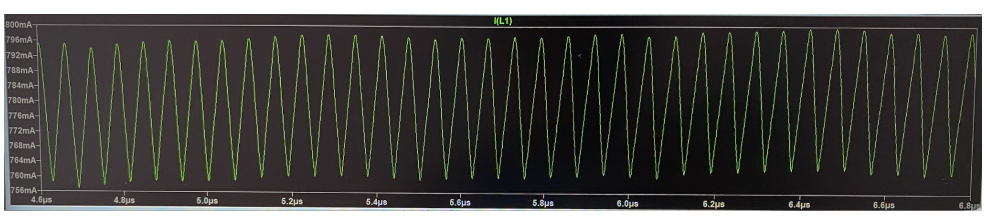


Figure B8. Shows drain current, used for deciding the value of C2.

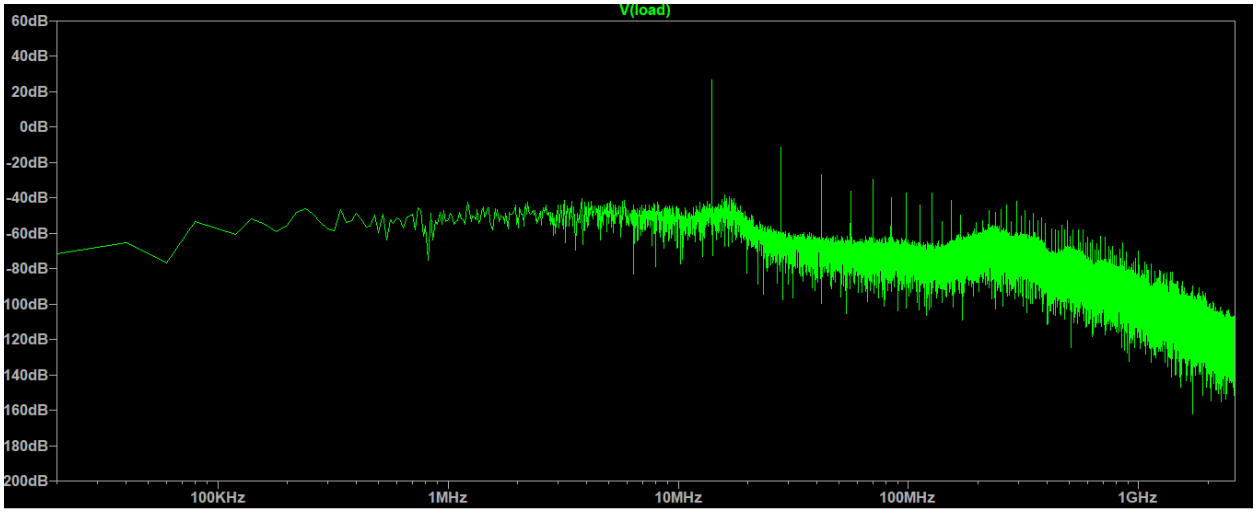


Figure B9. At 14 MHz we have 27dB. At 28 MHz (second harmonic) we have -9dB and it continues to drop for each subsequent harmonic.

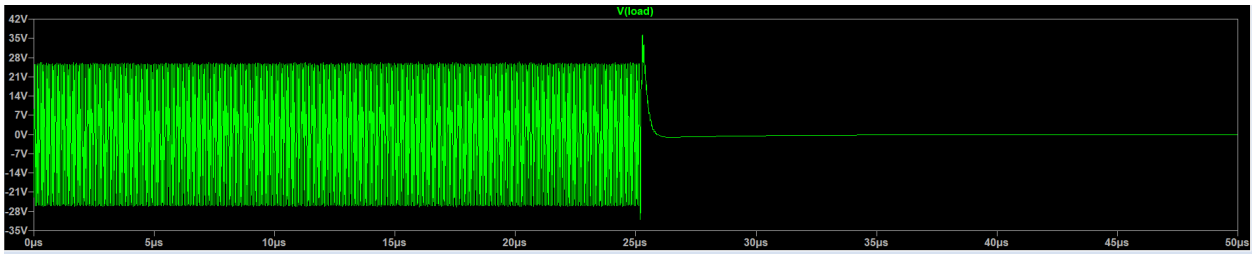


Figure B10. Shows that when TXEN becomes HIGH at 25us, the output voltage is approximately 0V.

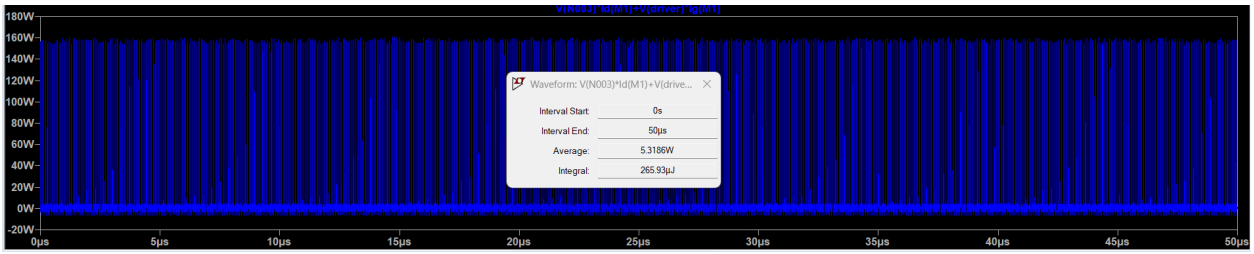
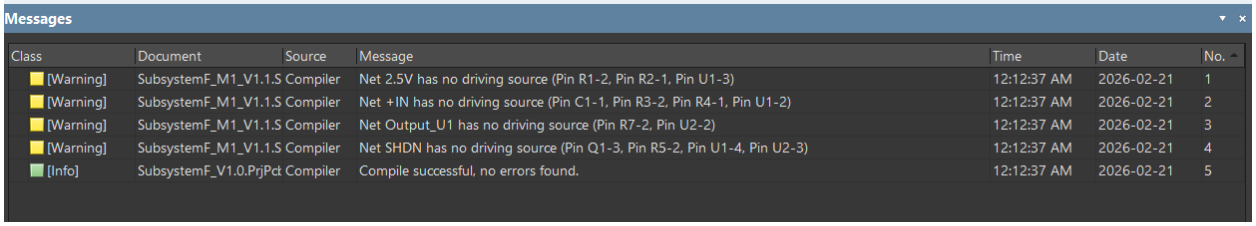


Figure B11. Shows the power dissipated by the MOSFET at 14 MHz.



Messages

Class	Document	Source	Message	Time	Date	No.
[Warning]	SubsystemF_M1_V1.1.S	Compiler	Net 2.5V has no driving source (Pin R1-2, Pin R2-1, Pin U1-3)	12:12:37 AM	2026-02-21	1
[Warning]	SubsystemF_M1_V1.1.S	Compiler	Net +IN has no driving source (Pin C1-1, Pin R3-2, Pin R4-1, Pin U1-2)	12:12:37 AM	2026-02-21	2
[Warning]	SubsystemF_M1_V1.1.S	Compiler	Net Output_U1 has no driving source (Pin R7-2, Pin U2-2)	12:12:37 AM	2026-02-21	3
[Warning]	SubsystemF_M1_V1.1.S	Compiler	Net SHDN has no driving source (Pin Q1-3, Pin R5-2, Pin U1-4, Pin U2-3)	12:12:37 AM	2026-02-21	4
[Info]	SubsystemF_V1.0.PrjPcb	Compiler	Compile successful, no errors found.	12:12:37 AM	2026-02-21	5

Figure B11. ERC check.

Appendix C: List of Components

Ref.	Component	Function / Value
U1	LTC6752-2	Ultrafast comparator / voltage driver, 5 V supply
U2	MCP14A0602	6 A peak gate driver / current driver, 12 V supply
Q2	IRF510	N-ch power MOSFET, Class D PA (TO-220, heatsink)
Q1	2N3904	NPN BJT, TXEN enable/disable gating
L1	10 µH	RF choke - drain bias feed for AC-coupled PA
L2	360 nH	1st series inductor, 5th-order Butterworth LPF
L3	680 nH	2nd series inductor, 5th-order Butterworth LPF
C1	0.1 µF	AC coupling, input from Subsystem E
C2	220 pF	RF bypass capacitor near Q2
C3	10 µF	12 V supply bypass capacitor
C4	100 nF	AC coupling, PA drain to output filter
C5	240 pF	First shunt capacitor, LPF (adjusted for IRF510 $C_{oss} \approx$ 140 pF)
C6	470 pF	Middle shunt capacitor, LPF impedance matching
C7	100 pF	Final shunt capacitor, load-side impedance smoothing
R1, R2	10 kΩ each	Voltage divider, 2.5 V comparator -IN reference bias
R3, R4	10 kΩ each	Voltage divider, 2.5 V comparator +IN bias
R5	10 kΩ	SHDN pull-up resistor to 5 V
R6	10 kΩ	Q1 base resistor from /TXEN
R7	1.5 Ω	Gate series resistor - damps switching ringing
R8	10 kΩ	Gate pull-down resistor for IRF510 (Q2)

Table C1. Key component summary.